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# Estimating Heterogeneous Causal Effect on Networks via Orthogonal Learning

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Anonymous Author(s)

Affiliation

Address

email

## Abstract

1 Estimating causal effects on networks is important for both scientific research and  
2 practical applications. Unlike traditional settings that assume the Stable Unit Treat-  
3 ment Value Assumption (SUTVA), interference allows an intervention/treatment on  
4 one unit to affect the outcomes of others. Understanding both *direct* and *spillover*  
5 effects is critical in fields such as epidemiology, political science, and economics.  
6 Causal inference on networks faces two main challenges. First, causal effects are  
7 typically heterogeneous, varying with unit features and local network structure.  
8 Second, connected units often exhibit dependence due to network homophily, cre-  
9 ating confounding between structural correlations and causal effects. In this paper,  
10 we propose a two-stage method to estimate heterogeneous direct and spillover  
11 effects on networks. The first stage uses graph neural networks to estimate nui-  
12 sance components that depend on the complex network topology. In the second  
13 stage, we adjust for network confounding using these estimates and infer causal  
14 effects through a novel attention-based interference model. Our approach balances  
15 expressiveness and interpretability, enabling downstream tasks such as identifying  
16 influential neighborhoods and recovering the sign of spillover effects. We integrate  
17 the two stages using Neyman orthogonalization and cross-fitting, which ensures  
18 that errors from nuisance estimation contribute only at higher order. As a result, our  
19 causal effect estimates are robust to bias and misspecification in modeling causal  
20 effects under network dependencies.

## 21 1 Introduction

22 Understanding causal effects on networks requires quantifying whether, and to what extent, a unit’s  
23 behavior is influenced by its interactions with others. The goal is to identify and measure the causal  
24 impact of network interference on individual outcomes. While modern data collection and social  
25 media provide large-scale network data with individual-level features, the complex structure of such  
26 data poses major challenges for causal inference. Traditional methods, which rely on independence  
27 assumptions, inevitably fail to capture the intricate dependencies present in real-world networks.

28 One of the major challenges is the inherent heterogeneity of causal effects under network interference.  
29 In real-world networks, both node features and the strengths of pairwise connections vary widely, so  
30 the way one unit’s outcome responds to its neighbors depends on a complex mix of individual traits  
31 and relational structure. Accurately capturing spillover heterogeneity is essential for identifying true  
32 causal effects and for developing theories that reflect diverse patterns of interaction across networks.

33 Another key challenge is the complex confounding introduced by the network structure. Units  
34 connected within a network exhibit correlated behaviors, driven not only by target interference but  
35 also by latent dependencies stemming from shared traits and network topology. Separating spillover

effect from non-causal associations arising from network topology requires sophisticated adjustments to handle high-dimensional features and latent relationships.

**Running example: political polarization.** Social media campaigns often utilize targeted ads (treatment) designed to influence voter turnout (outcome) [6]. These campaigns can *directly* influence recipients of targeted ads by encouraging them to vote, and also generate *spillover* effects as the voting message is reshared via social networks.

The magnitude of these spillover effects vary among voters depending on their ideological alignment and socioeconomic status. Critically, the sign of the spillover effects also differ based on voters’ ideological positions, which is the well-documented phenomenon of *political polarization and echo chambers* [3]. Moreover, political attitudes often leads to clusters of voters with similar turnout patterns in social media, i.e., *network homophily*.

Therefore, targeted ads exposure may be overlapped with groups of active voters, making it difficult to separate the causal impact of ads exposure from non-exposure voting patterns. The causal relation among outcome, treatment, and individual features over network is illustrated in Figure 1.

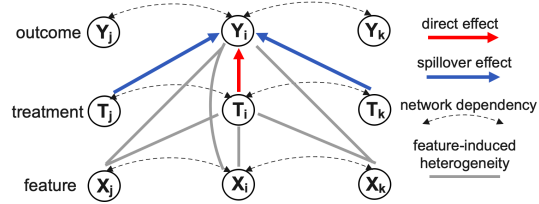


Figure 1: Causal diagram for an ego unit  $i$  on network where units  $j$  and  $k$  are two neighbors of  $i$ .

In this paper, we propose a novel method for estimating heterogeneous causal effects on networks from observational data under complex network dependencies. We introduce a new orthogonal learning design that separates the estimation of nuisance parameters arising from network dependencies from the estimation of causal effects, including both direct and spillover effects. The proposed method enjoys a robust estimation property: as long as the nuisance estimation bias is moderate, the estimation error for the causal effect remains of the same order as that of an oracle estimator with known nuisance components. In addition, the proposed framework balances expressive power and interpretability in modeling interference heterogeneity. Specifically, we use graph neural networks to estimate nuisance components, including both the conditional mean and propensity scores, therefore avoiding the strong parametric assumptions common in traditional methods. Inspired by attention mechanisms, our method introduces an attention-based interference model that summarizes the spillovers a unit receives from its neighborhood. This formulation allows spillover effects to depend on the features of both the sender and receiver, as well as on their local interaction patterns within the network.

## 2 Related works

Estimating heterogeneous causal effects in the presence of high-dimensional features and complex confounding has received increasing attention in recent years. Many methods have been developed within the frameworks of semiparametric theory [21] and double machine learning [8], primarily for treatment effect estimation from independent observational samples [10, 22, 20, 24, 32, 36]. Meanwhile, the widespread applications and theoretical importance of causal inference on networks have sparked a surge of interest in developing network-specific methodologies. However, the methods above are not directly applicable to network data due to interference between units, which violates the core independence assumption underlying classical causal inference frameworks [35]. Causal inference on networks introduces a distinct estimand—often referred to as the spillover effect, peer effect, or herd effect—that quantifies the causal influence transmitted through network connections. To address this, several recent methods have been proposed for causal effect estimation on networks [9, 1, 33, 12, 38, 26, 25]. A central component of model-based approaches is the use of exposure mappings, which summarize how a unit is affected by the treatment assignments of others [1]. Many existing methods assume a known exposure mapping [33, 17, 23, 7] or use simplified forms, such as the average treatment among neighbors [27]. These assumptions may be infeasible or overly simplistic in real-world settings, limiting the ability to capture complex interference mechanisms. Other methods aim to relax outcome model assumptions while requiring strong structural constraints, such as known clusters in the network [4, 16]. At the other end of the spectrum, deep learning-based approaches have emerged, where both interference and contextual information are encoded as latent representations in outcome models [28, 15, 29, 42]. However, the black-box nature of these models

90 makes it difficult to define interpretable treatment and spillover effects. More importantly, existing  
91 methods struggle to capture spillover effects between *pairs* of units connected on network.

### 92 3 Notations

93 Consider a network with  $n$  units  $\mathbf{V} = \{1, 2, \dots, n\}$ . Let  $\mathbf{t} = (t_1, \dots, t_n) \in \{0, 1\}^n$  denote  
94 the treatment assignment vector, where  $t_i$  is the treatment received by unit  $i$ . Under network  
95 interference, the potential outcome of unit  $i$  is denoted by  $Y_i(\mathbf{t})$ , highlighting its dependence on the  
96 entire treatment vector  $\mathbf{t}$ , including both  $t_i$  and the treatments of other units  $\mathbf{t}_{-i} = \mathbf{t} \setminus \{t_i\}$ . We  
97 observe unit-level features  $\mathbf{X} \in \mathbb{R}^{n \times p} = (X_1^\top, \dots, X_n^\top)^\top$ , where each  $X_i \in \mathcal{X}$  is a  $p$ -dimensional  
98 feature vector. The network is represented by an adjacency matrix  $\mathbf{A} \in \{0, 1\}^{n \times n}$ . Let  $l(i, j)$   
99 denote the shortest path length between units  $i$  and  $j$  in  $\mathbf{A}$ . We define the  $K$ -order neighborhood  
100 of unit  $i$  (including  $i$  itself) as  $\mathcal{N}_K(i) = \{j \in \mathbf{V} : l(i, j) \leq K\}$ , so that  $i \in \mathcal{N}_K(i)$ . The  
101 corresponding  $K$ -degree is  $d_K(i) = |\mathcal{N}_K(i)|$ . We denote the treatments and features within  $\mathcal{N}_K(i)$   
102 by  $\mathbf{t}_{\mathcal{N}_K(i)} = \{t_j : j \in \mathcal{N}_K(i)\}$  and  $\mathbf{X}_{\mathcal{N}_K(i)} = \{X_j : j \in \mathcal{N}_K(i)\}$ , respectively. Let  $(T_i, Y_i)_{i=1}^n$   
103 be the observed treatments and outcomes, and define  $\mathbf{T} = \{T_i\}_{i=1}^n$ ,  $\mathbf{Y} = \{Y_i\}_{i=1}^n$ . We define the  
104  $L_p$  norm for  $p \geq 1$  as  $\|f\|_p = (\mathbb{E}_D[|f(X)|^p])^{1/p}$ , where  $D$  is the distribution of  $X$ . We write  
105  $f(x) = \mathcal{O}(g(x))$  if there exists  $x_0$  such that  $|f(x)/g(x)| \leq M$  for all  $x > x_0$ .

### 106 4 Orthogonal heterogeneous interference structure

107 The goal is to infer causal estimands of network interference from observational data  $(T_i, Y_i)_{i=1}^n$ .  
108 We first introduce the individual total treatment effect (ITE) as  $\text{ITE}_i := \mathbb{E}(Y_i(\mathbf{t}) - Y_i(\mathbf{0}) \mid \mathbf{X}, \mathbf{A})$ .  
109 This total effect can be further decomposed into individual direct effect (IDE) as  $\text{IDE}_i := \mathbb{E}(Y_i(t_i =$   
110  $1, \mathbf{t}_{-i} = \mathbf{0}) - Y_i(t_i = 0, \mathbf{t}_{-i} = \mathbf{0}) \mid \mathbf{X}, \mathbf{A})$  and individual spillover effect (ISE) as  $\text{ISE}_i :=$   
111  $\mathbb{E}(Y_i(t_i = 0, \mathbf{t}_{-i} = \mathbf{1}) - Y_i(t_i = 0, \mathbf{t}_{-i} = \mathbf{0}) \mid \mathbf{X}, \mathbf{A})$ . All three estimands (ITE, IDE, and ISE) are  
112 node-level quantities that depend on each node's features and position in the network.

113 Without further assumptions, the causal quantities defined above are not informative due to the missing  
114 counterfactual realization of  $\mathbf{t}$ , are infeasible to estimate because of their complex dependence on  $\mathbf{X}$   
115 and  $\mathbf{A}$ . Therefore, we impose the following assumption on network interference:

116 **Assumption 1 (local interference):** The unit-wise conditional total treatment effect satisfies  
117  $\mathbb{E}(Y_i(\mathbf{t}) \mid \mathbf{X}, \mathbf{A}) = \mathbb{E}(Y_i(\mathbf{t}_{\mathcal{N}_1(i)}) \mid \mathbf{X}_{\mathcal{N}_1(i)}, \mathbf{A})$  for  $i = 1, \dots, n$

118 Assumption 1 restricts the structure of network interference to the one-hop neighborhood of unit  $i$ .  
119 As an alternative to the commonly used exposure mapping assumption [31, 2], the local interference  
120 assumption has also been considered in [26], aiming to create approximately independent interference  
121 samples from a single observation and to control the estimation complexity of causal estimands. Note  
122 that Assumption 1 does not rule out non-causal associations among outcomes of connected units in  
123 the network. To capture the heterogeneity of network interference and facilitate interpretation, we  
124 impose the following interference structure:

125 **Assumption 2 (Additive Network Interference):** For node  $i$ , and each neighbor  $j \in \mathcal{N}_1(i)$ , there  
126 exists a set of functions  $\{g_j(t_j, \mathbf{X}_{\mathcal{N}_1(i)}, \mathbf{A})\}_{j=1}^{d_1(i)}$  such that

$$\mathbb{E}(Y_i(\mathbf{t}_{\mathcal{N}_1(i)}) \mid \mathbf{X}_{\mathcal{N}_1(i)}, \mathbf{A}) = \sum_{j \in \mathcal{N}_1(i)} g_j(t_j, \mathbf{X}_{\mathcal{N}_1(i)}, \mathbf{A}) \quad (1)$$

127 Assumption 2 is a generalization of, and weaker than, the additive structure assumptions used in  
128 [14, 12, 30]. Notably, Assumption 2 allows for interactions among the interference effects from  
129 different units in  $\mathcal{N}_1(i)$ , since each component  $g_j(t_j, \mathbf{X}_{\mathcal{N}_1(i)}, \mathbf{A})$  can depend on the features of other  
130 neighbors and the network structure  $\mathbf{A}$ . To identify causal estimands from observational data, we  
131 introduce two standard assumptions below.

132 **Assumption 3 (unconfoundedness and positivity):** For node  $i$ ,  $Y_i(\mathbf{t}) \perp\!\!\!\perp (T_j)_{j \in \mathcal{N}_1(i)} \mid \mathbf{X}_{\mathcal{N}_1(i)}, \mathbf{A}$ ,  
133 and  $0 < P(T_j = 1 \mid \mathbf{X}, \mathbf{A}) < 1$ .

134 Assumption 3 can hold even in the presence of dependencies among  $T_j$  for  $j \in \mathcal{N}_1(i)$ , or among po-  
135 tential outcomes  $Y_j(\mathbf{t})$  in the same neighborhood. Unlike the traditional positivity assumption, which

requires support over the joint distribution of treatment assignments  $\{T_j\}_{j \in \mathcal{N}_1(i)}$  [9], Assumption 3 only requires positivity of the marginal propensity for each  $T_j$ .

**Theorem 1.** For unit  $i$ , define  $\tau_{ij} := g_j(1, \mathbf{X}_{\mathcal{N}_1(i)}, \mathbf{A}) - g_j(0, \mathbf{X}_{\mathcal{N}_1(i)}, \mathbf{A})$  for each  $j \in \mathcal{N}_1(i)$ . Under Assumptions 1 to 3, the unit-level ITE, IDE, and ISE are identifiable from the observed data  $(T_i, Y_i)_{i=1}^n$  via the conditional expectation  $\mathbf{E}(Y_i \mid \mathbf{t}_{\mathcal{N}_1(i)} = \mathbf{T}_{\mathcal{N}_1(i)}, \mathbf{X}_{\mathcal{N}_1(i)}, \mathbf{A})$ . Specifically,

$$\text{ITE}_i = \sum_{j \in \mathcal{N}_1(i)} \tau_{ij}, \quad \text{IDE}_i = \tau_{ii}, \quad \text{ISE}_i = \sum_{j \in \mathcal{N}_1(i), j \neq i} \tau_{ij}.$$

Furthermore, for any two treatment assignments  $\mathbf{t}$  and  $\mathbf{t}'$ , we have:

$$\mathbf{E}(Y_i(\mathbf{t}) - Y_i(\mathbf{t}') \mid \mathbf{X}, \mathbf{A}) = \sum_{j \in \mathcal{N}_1(i)} \tau_{ij}(t_j - t'_j).$$

#### 4.1 Orthogonal learning for network interference

Let  $\mathbf{\Gamma} = \{\tau_{ij}\} \in \mathbb{R}^{n \times n}$  denote the matrix of interference coefficients. Theorem 1 implies that the causal estimands of interest can be expressed as linear combinations of elements in  $\mathbf{\Gamma}$ . Under Assumptions 1 and 2, the outcome model can be rewritten as follows:

$$Y_i = \mathbf{E}(Y_i(0) \mid \mathbf{X}_{\mathcal{N}_1(i)}, \mathbf{A}) + \sum_{j \in \mathcal{N}_1(i)} T_j \tau_{ij} + \epsilon_i, \quad (2)$$

Here, the error term satisfies  $\mathbb{E}(\epsilon_i \mid \mathbf{T}_{\mathcal{N}_1(i)}, \mathbf{X}_{\mathcal{N}_2(i)}, \mathbf{A}) = 0$  under Assumption 3. To derive the orthogonal learning formulation for estimating  $\mathbf{\Gamma}$ , we first take the expectation of both sides of equation (2) with respect to  $(\mathbf{T}, \mathbf{Y})$ , conditional on  $\mathbf{X}$  and  $\mathbf{A}$ ,

$$Y_i - \mathbf{E}(Y_i \mid \mathbf{X}, \mathbf{A}) = \sum_{j \in \mathcal{N}_1(i)} (T_j - \mathbf{P}(T_j = 1 \mid \mathbf{X}, \mathbf{A})) \tau_{ij} + \epsilon_i. \quad (3)$$

The orthogonal learning formulation in equation (3) separates the estimation of nuisance components—specifically, the conditional outcome mean  $m := \mathbb{E}(Y \mid \mathbf{X}, \mathbf{A})$  and the propensity score  $e := \mathbb{P}(T = 1 \mid \mathbf{X}, \mathbf{A})$ —from the estimation of  $\mathbf{\Gamma}$ . These nuisance functions are relevant for identifying the target causal estimands, but errors in estimating  $m$  and  $e$  can introduce bias when estimating  $\mathbf{\Gamma}$ . The orthogonal loss design in (3) generalizes the R-Learner framework [32] and enables estimation of  $\mathbf{\Gamma}$  with generalization error comparable to that of an oracle estimator with known  $m$  and  $e$ . Given that  $m$  and  $e$  may have complex dependencies on  $\mathbf{X}$  and  $\mathbf{A}$ , we propose using expressive graph neural networks to estimate  $\hat{m}$  and  $\hat{e}$ , which are then plugged into equation (3) to estimate  $\mathbf{\Gamma}$ . An overview of the proposed framework for network interference is illustrated in Figure 2.

Orthogonal learning also relies on a cross-fitting procedure, where nuisance and target components are estimated on two independent subsets of the data to avoid overfitting bias. Unlike the i.i.d. setting, observations  $\{T_i, Y_i\}_{i=1}^n$  are typically dependent due to the network structure. This necessitates a refined cross-fitting strategy along with additional assumptions.

**Assumption 4:** (1)  $X_i \perp\!\!\!\perp X_j$  and  $T_i \perp\!\!\!\perp T_j$  if  $\mathbf{A}_{ij} = 0$ ; (2)  $Y_i \perp\!\!\!\perp Y_j$  if  $j \notin \mathcal{N}_3(i)$ .

Assumption 4 imposes constraints on the scope of network dependence, enabling estimation from a single snapshot of the network. Similar assumptions have been used in [33, 25] to ensure consistent ATE estimation. However, Assumption 4 is weaker, as it accommodates both local dependencies and long-range dependencies induced by the network structure  $\mathbf{A}$  and shared covariates  $\mathbf{X}$ . Given these preliminaries, we now present the following two-step estimators using cross-fitting:

Data Splitting: Randomly select a subset of units  $\mathbf{V}_1 \subset \mathbf{V}$ , and define

$$\mathbf{V}_2 = \bigcap_{i \in \mathbf{V}_1} \{j \notin \mathcal{N}_2(i)\}.$$

Introduce the set  $\tilde{\mathbf{V}}_s = \{j \in \mathcal{N}_1(i) \mid i \in \mathbf{V}_s\}$ ,  $s = 1, 2$ .

Stage I: Train and tune GNNs to obtain  $\hat{m}^{(s)}$  and  $\hat{e}^{(s)}$  based on the data folds  $(\{Y_i\}_{i \in \mathbf{V}_s}, \mathbf{X}, \mathbf{A})$  and  $(\{T_i\}_{i \in \tilde{\mathbf{V}}_s}, \mathbf{X}, \mathbf{A})$ , for  $s = 1, 2$ .

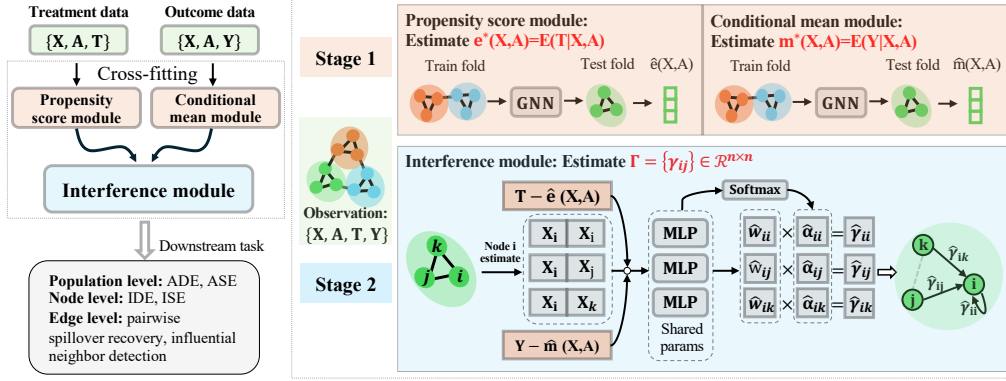


Figure 2: Two-stage orthogonal learning framework for estimating direct and spillover effects under an additive structure.

172 *Stage 2:* For each unit  $i$ , let  $-s_i \in \{1, 2\}$  denote the data fold that  $i$  does **not** belong to. Estimate the  
 173 interference coefficients by solving:

$$\hat{\Gamma} = \arg \min_{\Gamma = \{\tau_{ij}\}} \sum_{i \in V_1 \cup V_2} \left( Y_i - \hat{m}_i^{-s_i} - \sum_{j \in \mathcal{N}_1(i)} (T_j - \hat{e}_j^{-s_i}) \tau_{ij} \right)^2, \quad (4)$$

174 where  $\hat{m}_i^{-s_i}$  and  $\hat{e}_j^{-s_i}$  are the predictions of the conditional outcome mean and propensity score based  
 175 on GNNs trained on the opposite fold  $-s_i$ .

## 176 4.2 Attention-based network interference modeling

177 The interference matrix  $\Gamma$  is expected to capture: 1) the heterogeneity among  $\tau_{ij}$ , which depends  
 178 on features of both the focal node and its neighbors, and 2) interactions among neighboring units'  
 179 influence. To model these properties, we propose an attention-based interference model inspired by  
 180 the graph attention mechanism [40]. Specifically, for each unit  $i \in V$  and neighbor  $j \in \mathcal{N}_1(i)$ , we  
 181 define:

$$\tau_{ij} = \alpha_{ij} \times w_{ij}, \quad w_{ij} := W(X_i, X_j), \quad \alpha_{ij} := \alpha(w_{ij}, \mathbf{w}_i) \text{ s.t. } \alpha_{ij} \geq 0, \quad \sum_{j \in \mathcal{N}_1(i)} \alpha_{ij} = 1, \quad (5)$$

182 where  $\mathbf{w}_i$  is vector with element being  $w_{ij}$ . The bivariate function  $W(\cdot, \cdot)$  measures the influence  
 183 of unit  $j$  on unit  $i$ , and  $\alpha(\cdot)$  is a weighting function to aggregate neighborhood influence. In this  
 184 paper, we choose  $W = \text{MLP-ReLU} : \mathbb{R}^{2p} \rightarrow \mathbb{R}$ , a multi-layer ReLU neural network with  $\parallel$  being  
 185 the concatenation operation. We use a softmax function for  $\alpha$ , defined as  $\alpha_{ij} = (\text{softmax}(\beta \|\mathbf{w}_i\|))_j$ ,  
 186 where  $\beta > 0$  is a learnable temperature parameter. The softmax operation allows the model to flexibly  
 187 approximate various neighborhood aggregation schemes (e.g., maximum, minimum) by adjusting  
 188  $\beta$ . Model (5) captures both the sign and magnitude of interference heterogeneity, and it allows for  
 189 asymmetric influence ( $\tau_{ij} \neq \tau_{ji}$ ). Moreover, this attention-based formulation generalizes several  
 190 popular exposure mappings proposed in prior work [1].

## 191 4.3 Theoretical analysis

192 According to the interference model, causal effect estimation is primarily determined by the influence  
 193 function  $W$ . Therefore, we analyze the theoretical convergence of  $\hat{W}$  estimated via the orthogonal  
 194 learning method. Since the observations  $\{T_i, Y_i\}_{i=1}^n$  are not independent, we introduce the following  
 195 setup to study the empirical process on dependent data.

196 A dependence graph  $G \in \{0, 1\}^{n \times n}$  for a set of random variables  $V = \{V_i\}_{i=1}^n$  satisfies  $G_{ij} = 0$   
 197 if and only if  $V_i$  and  $V_j$  are independent. To quantify the intensity of dependence, we introduce  
 198 a *fractional independent cover*  $\mathcal{I}(G) := \{(I_k, \lambda_k)\}_{k=1}^J$ , where each  $I_k \subseteq V$  and  $\bigcup_k I_k = V$ .

In addition, all variable pairs within each set  $I_k$  are mutually independent (i.e., not adjacent in  $G$ ). Each subset  $I_k$  is associated with a weight  $\lambda_k \in [0, 1]$ , such that for every  $V_i \in V$ , the total weight satisfies  $\sum_{k: V_i \in I_k} \lambda_k = 1$ . The *fractional chromatic number* is then defined as  $\chi_f(G) := \min_{\mathcal{I}(G)} \sum_{I_k \in \mathcal{I}(G)} \lambda_k(I_k)$ . Next, consider the interference function class  $\mathcal{F} = \{W(X_i, X_j) : \mathbb{R}^p \times \mathbb{R}^p \rightarrow \mathbb{R} \mid |W(X_i, X_j)| \leq M, X_i, X_j \in \mathcal{X}\}$ . We define the *local fractional Rademacher complexity* of  $\mathcal{F}$  on dependent data over  $G$  as follows:

$$\mathcal{R}_n(\eta; \mathcal{F}, G) = \frac{1}{n} \sum_{k=1}^J \lambda_k \mathbb{E}_{\mathbf{X}, \boldsymbol{\sigma}} \left\{ \sup_{W \in \mathcal{F}: \|W - W^*\|_2 \leq \eta} \left| \sum_{i \in I_k} \sum_{j \in \mathcal{N}_1(i)} \sigma_{ij} W(X_i, X_j) \right| \right\}$$

where  $\{(I_k, \lambda_k)\}_{k=1}^J$  is a fractional independent cover satisfying  $\sum_{k=1}^J \lambda_k = \chi_f(G)$ , and  $\boldsymbol{\sigma} = \{\sigma_{ij}\}$  denotes a collection of independent Rademacher variables such that  $\mathbb{P}(\sigma_{ij} = 1) = \mathbb{P}(\sigma_{ij} = -1) = 1/2$ . Since  $\hat{W}$  depends on nuisance components estimated in the first stage, we introduce the following:

**Assumption 5:** With probability at least  $1 - \delta$ , the estimation of the conditional mean and propensity score satisfies the following property:

$$\|\hat{m} - m^*\|_4 \leq \eta_m, \|\hat{e} - e^*\|_4 \leq \eta_e, \quad (6)$$

Assumption 5 requires that the prediction errors of the nuisance estimators can be controlled. In our work, we use graph neural networks as the estimators, whose generalization properties have been studied in [11, 37, 39]. The  $L_4$  norm condition can be relaxed to an  $L_2$  norm under the assumption that  $\mathbf{X}$  follows a sub-Gaussian distribution.

**Theorem 2.** Under Assumption 5 and 6, and regularity conditions: 1)  $\sup_{W, m, e} |\hat{\mathbb{E}}(Y_i | \mathbf{X}, \mathbf{T}, \mathbf{A})| \leq L_1$ ; 2)  $\sup_{W, m, e} |l(W, m, e)| \leq L_2$  where  $l(\cdot)$  is the  $l_2$  loss (4); 3)  $\mathbb{P}(T_i = 1 | \mathbf{X}, \mathbf{A}) \in [c, 1 - c]$  where  $0 < c < 1/2$ . Consider the dependency graph  $G_A = \{0, 1\}^{n \times n}$  where  $G_{ij} = 1$  if  $j \in \mathcal{N}_3(i)$ . Let  $\eta_W$  be the solution to the equation:

$$\mathcal{R}_n(\eta_W; \mathcal{F}, G_A) \leq \eta_W^2,$$

If true  $W^* \in \mathcal{F}$ , then with probability larger than  $1 - c_n \exp\{-c_1 n \eta_W^4\} - 2\delta$ , we have

$$\|\hat{W} - W^*\|_2^2 = \mathcal{O}\left(\frac{L_2 L_1^2 d_{max}^2 \sqrt{\chi_f(G_A)}}{c^2(1 - c^2)} \eta_W^2 + \frac{M^2 d_{max}^4}{c^3(1 - c)^3} (\eta_m^4 + \eta_e^4)\right),$$

where  $d_{max} := \max_i \{|\sum_j \mathbf{A}_{ij}|\}$ ,  $c_1 > 0$  is constant, and  $c_n \rightarrow 0$  as  $n \rightarrow \infty$ . The fractional chromatic number can be bounded by maximum node degree [5], i.e.,  $\chi_f(G_A) \leq d_{max}^4$ .

Theorem 2 shows that the estimation error of  $\hat{W}$  depends on the nuisance estimation errors  $\eta_m$  and  $\eta_e$ , as well as the second-stage error  $\eta_W$ , which is computed based on  $\hat{m}$  and  $\hat{e}$ . Notably,  $\eta_m$  and  $\eta_e$  contribute only at the second order relative to  $\eta_W$ . Therefore, if  $\hat{m}$  and  $\hat{e}$  converge at a moderately faster rate compared to  $\hat{W}$ , specifically when  $\eta_m, \eta_e = \mathcal{O}(\eta_W^{1/2})$ , then  $\hat{W}$  achieves the same convergence rate as the oracle estimator that assumes  $m$  and  $e$  are known.

## 5 Experiment

In Section 5.1, we compare our proposed method against baselines on two benchmark network datasets for causal effect estimation. In Section 5.2, we further evaluate our method's ability to recover heterogeneous edge-level spillover effects and to generate interpretable insights for practical applications.

**Estimands and evaluation metrics.** We adopt causal estimands commonly used in real-world applications [42, 7]. At the individual level, we consider the Individual Direct Effect (IDE) and Individual Spillover Effect (ISE) as defined in Section 4. At the population level, we define the Average Direct Effect (ADE) and Average Spillover Effect (ASE) as  $\text{ADE} := \frac{1}{n} \sum_{i=1}^n \text{IDE}_i$  and  $\text{ASE} := \frac{1}{n} \sum_{i=1}^n \text{ISE}_i$ , respectively. To assess estimation accuracy, we use mean absolute error (MAE) for ADE and ASE, defined as  $|\hat{\tau} - \tau|$ , and precision in estimating heterogeneous effects (PEHE) for IDE and ISE, defined as  $\sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{\tau}_i - \tau_i)^2}$ .

**Sample splitting via graph partitioning.** Theoretically, the cross-fitting procedure requires two independent subsets of units,  $V_1$  and  $V_2$ , separated by a margin in graph distance. This involves discarding all responses  $\{Y_i\}$  located in between, potentially reducing training efficiency. In practice, we replace the margin split with a balanced graph partitioning step using METIS algorithm [19]. We specify  $S \geq 2$  roughly equal-sized parts  $\{V_s\}_{s=1}^S$  that minimize edge-cuts between clusters. In steps 2–3, we perform cross-fitting by training nuisance components on  $V_s$  and estimating the interference model on the complement  $\bigcup_{s' \neq s} V_{s'}$ , iterating through all subsets  $s$ . Data splitting via graph partitioning leverages all observed outcomes  $\{Y_i\}$  and ensures balanced sample sizes for nuisance and causal-effect estimation. Empirically, this approach scales effectively and provides stable performance in our experiments. We report results with  $S = 5$  in the following sections. Additional results with varying numbers of partitions are provided in the appendix.

## 5.1 Benchmark comparisons on real networks

**Data generation and setup.** Due to the common challenge of unobserved counterfactual treatments and outcomes in causal inference, we follow standard practice by evaluating our method in a semi-synthetic setting using two real social network datasets, BlogCatalog (BC) and Flickr, where node features  $\mathbf{X}$  and network structure  $\mathbf{A}$  are provided by the dataset. We then generate each node’s binary treatment  $T_i$  and outcome  $Y_i$  following [29, 7, 42]

$$T_i \sim \text{Bernoulli}(\sigma(f_T(\mathbf{X}_{\mathcal{N}_1(i)}))), \quad Y_i = f_0(\mathbf{X}_{\mathcal{N}_1(i)}) + \sum_{j \in \mathcal{N}_1(i)} f_1(T_j, W(X_i, X_j), \mathbf{X}_{\mathcal{N}_1(i)}) + \epsilon_i,$$

where  $f_T, f_1, f_2$  are summarization functions,  $\sigma(\cdot)$  is the sigmoid function and  $\epsilon_i$  is random noise. Details of the data-generation procedures for these functions appear in the Appendix. To reflect different interference patterns, we define  $W(\cdot)$  in three settings: (1) **Cosine** and **RBF**: pairwise kernel capturing heterogeneous interference based on both  $X_i$  and  $X_j$ ; (2) **One-way** non-interaction function defined as  $W(X_i, X_j) = f(X_j)$  where  $f(\cdot)$  is a non-linear function for  $j \in \mathcal{N}_1(i)$ . (3) **Homo** homogeneous interference where  $W(\cdot, \cdot)$  is constant. Settings (2) and (3) match the network causal models considered in [29, 7, 42], while setting (1) introduces a new heterogeneous and nonlinear models. In addition, we vary temperature  $\beta$  in attention weights  $\{\alpha_{ij}\}$  to mimic different local interference interaction, ranging from uniform spillovers ( $\beta = 0$ ) to sparse spillovers from influential neighbors ( $\beta = 10$ ). Details on the benchmark datasets, data generation procedures, model and hyperparameter setups are provided in the Appendix.

**Baselines.** For fair comparison, we consider seven baseline models that can estimate causal estimands at node-level (IDE, ISE) or population-level (ADE, ASE). **CFR** [18] is a widely-used neural network model for heterogeneous treatment effect estimation, and we adapt it to network data by incorporating neighborhood treatment and feature summaries as additional inputs. **NetEst** [17], **ND** [13], and **Caugamer** [42] are GNN-based methods that learn balanced node representations to control for network confounding in estimating causal effects. **GDML** [23] and **Tnet** [7] both construct doubly robust estimators for ADE and ASE. **EdgeConv** [41] is a graph convolutional neural network allows heterogeneous neighbor weights in message passing, and we adapt it for causal inference task. **EdgeConv** also serves as an ablation study where the causal estimation only consider interference heterogeneity without adjustments for network confounding.

**Evaluations.** Table 1 reports the out-of-sample estimation accuracy on the semi-synthetic Flickr dataset using **Cosine** and **RBF** kernels as interference models. The proposed estimator (**Proposed<sub>est</sub>**) performs competitively on population-level metrics (ADE and ASE), closely matching DBML and TNet. More notably, our method substantially improves node-level estimates (IDE and ISE), and improves increases as temperature  $\beta$  becoming larger, i.e., spillovers become sparse and concentrated on a few key neighbors. We also compare against the oracle estimator (**Proposed<sub>oracle</sub>**), which plugs in the true nuisance components (i.e., propensity scores and conditional means) while estimating causal effects. The consistently small gaps in performance metrics validate the effectiveness of our orthogonal learning design in mitigating nuisance estimation bias. Similar results on the BlogCatalog dataset (provided in the appendix) confirm that our method consistently delivers strong causal estimation performance at both population and individual levels.

Table 2 reports the causal estimation accuracy under the **One-way** and **Homogeneous** interference models on both Flickr and BlogCatalog networks. Our method consistently achieves state-of-the-art performance across most settings, highlighting the flexibility and robustness of our estimator under the varied interference patterns and network confounding structures common in existing literature.

Table 1: Causal estimation performance on the Flickr network using Cosine and RBF kernels as interference function with varying temperatures (Temps.)  $\beta \in \{0, 1, 5, 10\}$ . We highlight best performance in bold and the second-best with underline among the proposed method (**Proposed<sub>est</sub>**) and all baselines. We also report results of our method with known nuisance (**Proposed<sub>oracle</sub>**). IDE and ISE are measured by PEHE, whereas ADE and ASE are measured by MAE.

| Interference | Temp. | Effect | CFR          | EdgeConv             | ND           | Netest               | Tnet                 | Caugamer             | GDML                 | Proposed <sub>est</sub> | Proposed <sub>oracle</sub> |
|--------------|-------|--------|--------------|----------------------|--------------|----------------------|----------------------|----------------------|----------------------|-------------------------|----------------------------|
| Cosine       | 0     | ADE    | 0.1958±0.062 | 0.1715±0.034         | 0.2831±0.042 | 0.1975±0.063         | 0.1170±0.078         | 0.0955±0.093         | <b>0.0007</b> ±0.009 | 0.0120±0.006            | 0.0012±0.005               |
|              |       | ASE    | 0.0364±0.019 | 0.0867±0.015         | 0.0558±0.038 | 0.1396±0.047         | <b>0.0038</b> ±0.004 | 0.1073±0.090         | 0.0499±0.073         | 0.0046±0.009            | 0.0112±0.016               |
|              |       | IDE    | 0.3296±0.058 | 0.2603±0.039         | 0.3903±0.044 | 0.3274±0.061         | 0.3110±0.043         | 0.3202±0.052         | <b>0.0088</b> ±0.009 | 0.0277±0.008            | 0.0190±0.005               |
|              |       | ISE    | 0.2724±0.012 | <u>0.2102</u> ±0.005 | 0.3080±0.017 | 0.3190±0.022         | 0.3061±0.026         | 0.3121±0.035         | 0.2862±0.018         | <b>0.0409</b> ±0.004    | 0.0381±0.010               |
|              | 1     | ADE    | 0.2299±0.086 | 0.2051±0.031         | 0.2968±0.058 | 0.2101±0.074         | 0.1541±0.103         | 0.0812±0.103         | <u>0.0230</u> ±0.005 | <b>0.0080</b> ±0.005    | 0.0039±0.004               |
|              |       | ASE    | 0.0409±0.022 | 0.1259±0.020         | 0.0450±0.040 | 0.1221±0.056         | <b>0.0024</b> ±0.001 | 0.1658±0.109         | 0.0399±0.068         | <u>0.0041</u> ±0.013    | 0.0033±0.005               |
|              |       | IDE    | 0.3577±0.065 | 0.2868±0.034         | 0.4035±0.055 | 0.3374±0.064         | 0.3407±0.055         | 0.3404±0.040         | <u>0.0763</u> ±0.009 | <b>0.0252</b> ±0.008    | 0.0158±0.005               |
|              |       | ISE    | 0.2925±0.024 | <u>0.2405</u> ±0.010 | 0.3192±0.005 | 0.3498±0.024         | 0.3160±0.022         | 0.3541±0.048         | 0.2819±0.018         | <b>0.0499</b> ±0.005    | 0.0362±0.006               |
|              | 5     | ADE    | 0.1701±0.090 | 0.3342±0.036         | 0.1711±0.077 | 0.0896±0.069         | 0.2841±0.173         | <u>0.0758</u> ±0.059 | 0.1043±0.022         | <b>0.0028</b> ±0.002    | 0.0018±0.005               |
|              |       | ASE    | 0.1620±0.076 | 0.1431±0.048         | 0.1723±0.112 | 0.1182±0.080         | 0.0313±0.037         | 0.2039±0.177         | <u>0.0282</u> ±0.074 | <b>0.0012</b> ±0.005    | 0.0046±0.005               |
|              |       | IDE    | 0.3847±0.048 | 0.4297±0.032         | 0.3790±0.039 | 0.2999±0.037         | 0.4852±0.094         | 0.3922±0.127         | <u>0.2974</u> ±0.024 | <b>0.0289</b> ±0.005    | 0.0184±0.005               |
|              |       | ISE    | 0.4102±0.042 | 0.3093±0.030         | 0.4085±0.051 | 0.4462±0.022         | 0.3863±0.039         | 0.4334±0.083         | <u>0.2643</u> ±0.023 | <b>0.0298</b> ±0.004    | 0.0178±0.005               |
|              | 10    | ADE    | 0.1558±0.064 | 0.3398±0.044         | 0.1689±0.093 | 0.0785±0.047         | 0.3014±0.171         | <u>0.0249</u> ±0.028 | 0.1124±0.022         | <b>0.0063</b> ±0.008    | 0.0017±0.006               |
|              |       | ASE    | 0.1357±0.098 | 0.1450±0.054         | 0.1836±0.088 | 0.0906±0.064         | <u>0.0278</u> ±0.044 | 0.1875±0.166         | 0.0435±0.098         | <b>0.0028</b> ±0.004    | 0.0048±0.006               |
|              |       | IDE    | 0.3824±0.033 | 0.4382±0.048         | 0.3834±0.050 | <u>0.2945</u> ±0.016 | 0.4986±0.091         | 0.3106±0.047         | 0.3188±0.024         | <b>0.0293</b> ±0.009    | 0.0200±0.006               |
|              |       | ISE    | 0.4276±0.053 | 0.3091±0.020         | 0.4218±0.046 | 0.4384±0.015         | 0.3934±0.039         | 0.4467±0.064         | <u>0.2838</u> ±0.023 | <b>0.0317</b> ±0.003    | 0.0204±0.005               |
| RBF          | 0     | ADE    | 0.2116±0.069 | 0.1680±0.028         | 0.2758±0.077 | 0.2213±0.073         | 0.1142±0.076         | <u>0.0134</u> ±0.009 | <b>0.0013</b> ±0.004 | 0.0155±0.007            | 0.0007±0.006               |
|              |       | ASE    | 0.0410±0.035 | 0.1378±0.023         | 0.0442±0.029 | 0.1535±0.039         | <b>0.0032</b> ±0.002 | 0.1455±0.081         | 0.0506±0.060         | <u>0.0044</u> ±0.010    | 0.0067±0.011               |
|              |       | IDE    | 0.3360±0.063 | 0.2552±0.032         | 0.3834±0.076 | 0.3448±0.073         | 0.3095±0.044         | 0.3057±0.037         | <b>0.0034</b> ±0.005 | <u>0.0327</u> ±0.009    | 0.0164±0.005               |
|              |       | ISE    | 0.2287±0.013 | 0.2119±0.013         | 0.2439±0.011 | 0.3030±0.027         | 0.2492±0.021         | 0.2694±0.030         | <u>0.2111</u> ±0.022 | <b>0.0378</b> ±0.005    | 0.0281±0.008               |
|              | 1     | ADE    | 0.2188±0.072 | 0.2181±0.031         | 0.2623±0.040 | 0.2176±0.073         | 0.1720±0.111         | 0.0597±0.079         | <u>0.0339</u> ±0.008 | <b>0.0090</b> ±0.005    | 0.0058±0.005               |
|              |       | ASE    | 0.0626±0.038 | 0.1573±0.030         | 0.0814±0.049 | 0.1611±0.045         | 0.0097±0.020         | 0.1602±0.147         | 0.0289±0.052         | <b>0.0038</b> ±0.008    | 0.0018±0.014               |
|              |       | IDE    | 0.3535±0.061 | 0.2954±0.031         | 0.3793±0.044 | 0.3426±0.061         | 0.1708±0.033         | 0.3242±0.068         | <u>0.0595</u> ±0.015 | <b>0.0405</b> ±0.007    | 0.0281±0.004               |
|              |       | ISE    | 0.2903±0.015 | 0.2918±0.020         | 0.2755±0.021 | 0.3145±0.077         | <u>0.2478</u> ±0.005 | 0.3301±0.063         | 0.2871±0.060         | <b>0.0800</b> ±0.020    | 0.0575±0.008               |
|              | 5     | ADE    | 0.1701±0.090 | 0.3342±0.036         | 0.1711±0.077 | 0.0896±0.069         | 0.2841±0.173         | <u>0.0758</u> ±0.059 | 0.1043±0.022         | <b>0.0028</b> ±0.002    | 0.0018±0.005               |
|              |       | ASE    | 0.1620±0.076 | 0.1431±0.048         | 0.1723±0.112 | 0.1182±0.080         | 0.0313±0.037         | 0.2039±0.177         | <u>0.0282</u> ±0.074 | <b>0.0012</b> ±0.005    | 0.0046±0.005               |
|              |       | IDE    | 0.3847±0.048 | 0.4297±0.032         | 0.3790±0.039 | 0.2999±0.037         | 0.4852±0.094         | 0.3922±0.127         | <u>0.2974</u> ±0.024 | <b>0.0289</b> ±0.005    | 0.0184±0.005               |
|              |       | ISE    | 0.4102±0.042 | 0.3093±0.030         | 0.4085±0.051 | 0.4462±0.022         | 0.3863±0.039         | 0.4334±0.083         | <u>0.2643</u> ±0.023 | <b>0.0293</b> ±0.004    | 0.0178±0.005               |
|              | 10    | ADE    | 0.1558±0.064 | 0.3398±0.044         | 0.1689±0.093 | 0.0785±0.047         | 0.3014±0.171         | <u>0.0249</u> ±0.028 | 0.1124±0.022         | <b>0.0063</b> ±0.008    | 0.0017±0.006               |
|              |       | ASE    | 0.1357±0.098 | 0.1450±0.054         | 0.1836±0.088 | 0.0906±0.064         | <u>0.0278</u> ±0.044 | 0.1875±0.166         | 0.0435±0.098         | <b>0.0028</b> ±0.004    | 0.0048±0.006               |
|              |       | IDE    | 0.3824±0.033 | 0.4382±0.048         | 0.3834±0.050 | <u>0.2945</u> ±0.016 | 0.4986±0.091         | 0.3106±0.047         | 0.3188±0.024         | <b>0.0293</b> ±0.009    | 0.0200±0.006               |
|              |       | ISE    | 0.4276±0.053 | 0.3091±0.020         | 0.4218±0.046 | 0.4384±0.015         | 0.3934±0.039         | 0.4467±0.064         | <u>0.2838</u> ±0.023 | <b>0.0317</b> ±0.003    | 0.0204±0.005               |

Table 2: Causal estimation performance on the BC and Flickr networks using non-interaction and homogeneous interference function.

| Dataset | Interference | Effect | CFR                  | EdgeConv             | ND                   | Netest       | Tnet                 | Caugamer     | GDML                 | Proposed <sub>est</sub> | Proposed <sub>oracle</sub> |
|---------|--------------|--------|----------------------|----------------------|----------------------|--------------|----------------------|--------------|----------------------|-------------------------|----------------------------|
| BC      | One-way      | ADE    | 0.1254±0.094         | 0.0131±0.004         | 0.1426±0.081         | 0.0520±0.010 | <u>0.0015</u> ±0.002 | 0.0055±0.005 | <b>0.0007</b> ±0.015 | 0.0031±0.005            | 0.0018±0.004               |
|         |              | ASE    | 0.0901±0.060         | 0.1858±0.019         | 0.1493±0.111         | 0.2487±0.069 | <b>0.0072</b> ±0.006 | 0.1823±0.119 | 0.0721±0.077         | <u>0.0156</u> ±0.029    | 0.0020±0.017               |
|         |              | IDE    | 0.1940±0.066         | <u>0.0583</u> ±0.009 | 0.2230±0.058         | 0.1465±0.011 | 0.1346±0.006         | 0.1416±0.028 | 0.1190±0.013         | <b>0.0248</b> ±0.008    | 0.0198±0.006               |
|         |              | ISE    | 0.4016±0.035         | <u>0.3003</u> ±0.018 | 0.4307±0.066         | 0.4147±0.056 | 0.4224±0.016         | 0.4268±0.071 | 0.4096±0.025         | <b>0.0708</b> ±0.018    | 0.0610±0.008               |
|         | Homo         | ADE    | 0.1150±0.042         | 0.9119±0.019         | 0.1388±0.033         | 0.0235±0.019 | 0.4504±0.152         | 0.2553±0.072 | <u>0.0064</u> ±0.014 | <b>0.0002</b> ±0.007    | 0.0130±0.040               |
|         |              | ASE    | 0.0636±0.029         | 0.1499±0.042         | 0.1039±0.061         | 0.0925±0.031 | <u>0.0532</u> ±0.104 | 0.4226±0.076 | 0.1353±0.151         | <b>0.0031</b> ±0.013    | 0.0182±0.019               |
|         |              | IDE    | 0.1704±0.034         | 0.9166±0.017         | 0.1700±0.020         | 0.0969±0.007 | 0.5192±0.107         | 0.3076±0.060 | <b>0.0132</b> ±0.005 | <u>0.0352</u> ±0.011    | 0.0900±0.025               |
|         |              | ISE    | <u>0.1238</u> ±0.019 | 0.2528±0.020         | 0.1456±0.049         | 0.1319±0.025 | 0.1692±0.051         | 0.4253±0.072 | 0.1514±0.134         | <b>0.0385</b> ±0.008    | 0.0463±0.010               |
| Flickr  | One-way      | ADE    | 0.0287±0.023         | 0.0133±0.005         | 0.1078±0.040         | 0.0515±0.010 | <u>0.0028</u> ±0.002 | 0.0451±0.064 | 0.0152±0.019         | <b>0.0010</b> ±0.005    | 0.0011±0.005               |
|         |              | ASE    | 0.0989±0.061         | 0.0109±0.006         | 0.0956±0.065         | 0.0735±0.031 | <u>0.0042</u> ±0.002 | 0.0509±0.042 | 0.0183±0.040         | <b>0.0036</b> ±0.016    | 0.0024±0.008               |
|         |              | IDE    | 0.2346±0.045         | <u>0.1391</u> ±0.028 | 0.2940±0.028         | 0.2595±0.051 | 0.2630±0.041         | 0.2654±0.043 | 0.2454±0.048         | <b>0.0201</b> ±0.008    | 0.0135±0.004               |
|         |              | ISE    | 0.2888±0.031         | <u>0.1665</u> ±0.012 | 0.2801±0.034         | 0.2219±0.015 | 0.2693±0.012         | 0.2562±0.021 | 0.2544±0.016         | <b>0.0387</b> ±0.008    | 0.0257±0.003               |
|         | Homo         | ADE    | 0.0801±0.012         | 0.7799±0.075         | 0.1662±0.018         | 0.1299±0.019 | 0.6765±0.123         | 0.1426±0.089 | <u>0.0092</u> ±0.022 | <b>0.0001</b> ±0.003    | 0.0011±0.016               |
|         |              | ASE    | 0.0586±0.018         | 0.0775±0.025         | 0.0695±0.037         | 0.1699±0.039 | <u>0.0212</u> ±0.031 | 0.2481±0.139 | 0.1674±0.135         | <b>0.0053</b> ±0.010    | 0.0049±0.017               |
|         |              | IDE    | 0.1565±0.025         | 0.8327±0.044         | 0.1882±0.016         | 0.1656±0.017 | 0.6988±0.118         | 0.2634±0.120 | <u>0.0145</u> ±0.018 | <b>0.0087</b> ±0.002    | 0.0537±0.022               |
|         |              | ISE    | 0.1780±0.025         | 0.2462±0.042         | <u>0.1736</u> ±0.037 | 0.2348±0.031 | 0.2421±0.051         | 0.3011±0.111 | 0.2296±0.099         | <b>0.0316</b> ±0.014    | 0.0490±0.013               |

## 5.2 Causal estimation interpretability

In addition to compare on benchmark metrics, we demonstrate our method’s advantages for enhancing the interpretability of causal effects using synthetic data. Specifically, we evaluate our method’s ability to (1) recover heterogeneous patterns of neighborhood influence, (2) identify influential neighbors, and (3) distinguish between positive and negative spillovers. Motivated by the *political polarization* example introduced in Section 1, we simulate a network of  $n = 7500$  nodes with community structure, reflecting groups of different political ideology.

Each node  $i$  is then assigned a  $d$ -dimensional feature drawn from a community-specific distribution. The pairwise influence  $\{w_{ij}\}$  are built on node embeddings such that within-community and between-community  $\{w_{ij}\}$  have varying intensity and signs. This setup mimics realistic political polarization scenarios [3].

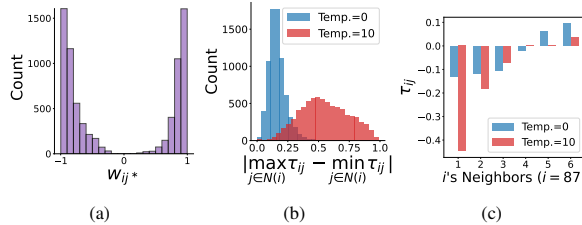


Figure 3: Spillover effects in political polarization

Figure 3a shows the distribution of simulated  $\{w_{ij^*}\}$ , where  $j^* = \arg \max_{j \in \mathcal{N}(i)} |w_{ij}|$ , revealing the groups of nodes whose most influential neighbour has opposite sign. As the temperature  $\beta$  increases, Figure (3b) shows a widening gap between a node’s strongest and weakest neighbor influences, reflecting greater heterogeneity in neighbor influences. Figure 3c examines the neighbor interference structure of example node  $i = 87$  at  $\beta = 0$  and  $\beta = 1$ . Further details are provided in the Appendix.

**Pairwise influence recovery.** We group raw pairwise influence  $\{w_{ij}\}$  into five groups based on their signs and magnitudes to reflect different ideological interaction strength. Figure 4a compares the group-specific distributions of (1) true  $w_{ij}$  (**True**), (2) estimated  $\hat{w}_{ij}$  using the true nuisance parameters  $m^*, e^*$  (**Oracle**), and (3) estimated  $\hat{w}_{ij}$  using cross-fitted GNN-based nuisance estimators  $\hat{m}, \hat{e}$  (**GNN**). Our method accurately identifies five categories distinguished by intensity and sign of pairwise influence. Within each category, our estimator closely matches the true medians and quantiles of  $\{w_{ij}\}$ , while exhibiting slightly larger variance due to expected nuisance-estimation noise.

**Influential neighbors detection.** For each ego node  $i$ , we evaluate our method’s performance in identifying its influential neighbors, defined as  $i$ ’s top 20% of neighbors ranked by the magnitude of their interference coefficients  $\{|\tau_{ij}|\}_{j \in \mathcal{N}_1(i)}$ . We measure performance using two complementary metrics: (1) **Recall@K=20%**: how well the model’s predicted top 20% neighbors *cover* the true top 20%; (2) **NDCG@K=20%**: how well the model *ranks* its predicted top 20% neighbors by rewarding higher placement of neighbors with larger true  $\{|\tau_{ij}|\}$ . The formal definitions of the two metrics are in Appendix. In addition to Oracle and GNN, we include **EdgeConv** as a baseline to illustrate performance without adjusting for network confounding. Figure 4b and 4c show that our method consistently recovers over 80% of the true top-20% neighbors of ego nodes on average, and perform closely to oracle estimation. Our method improves slightly with increasing temperature  $\beta$  due to that higher  $\beta$  amplifies the magnitude of spillovers from top neighbors hence making them easier to identify. In contrast, EdgeConv performs substantially worse.

**Spillover sign recovery.** We evaluate the performance in recovering the spillover signs for each node  $i$ ’s one-hop neighbors using the metric  $\frac{1}{|\mathcal{N}_1(i)|} \sum_{j \in \mathcal{N}_1(i)} \mathbf{1}[\text{sign}(\hat{w}_{ij}) = \text{sign}(w_{ij})]$ . Figure 4d shows the average estimation accuracy across all nodes. Our method achieves near-perfect recovery, significantly outperforming EdgeConv.

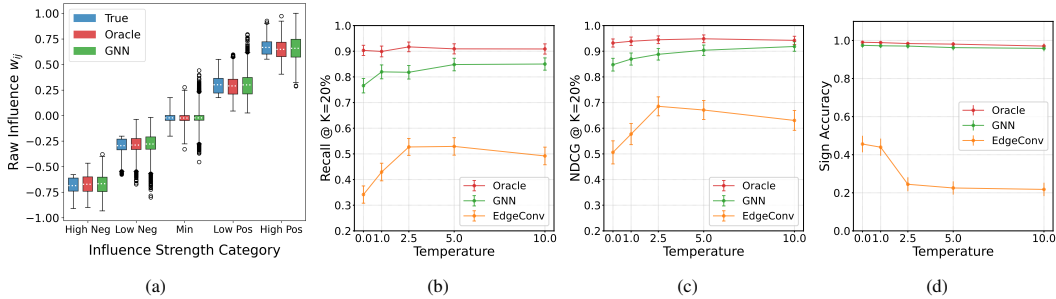


Figure 4: Edge-level interference estimation. (a): pairwise influence recovery. (b,c): influential neighbors detection. (d): spillover sign recovery.

## 6 Conclusion

In this paper, we propose a framework for estimating heterogeneous causal effects on network data. Our method is robust to bias in nuisance function estimation with theoretical guarantees. Moreover, the model balances interpretability and flexibility, supporting a variety of downstream analyses. Although our experiments offer a more generalizable and comprehensive evaluation than prior work, we acknowledge the limitation of relying on semi-synthetic datasets, which is a common challenge in causal inference due to the absence of counterfactuals. As future directions, we plan to adapt model-agnostic generative frameworks for validating causal inference methods [34] to network settings, and to further generalize our framework for more flexible aggregation schemes.

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